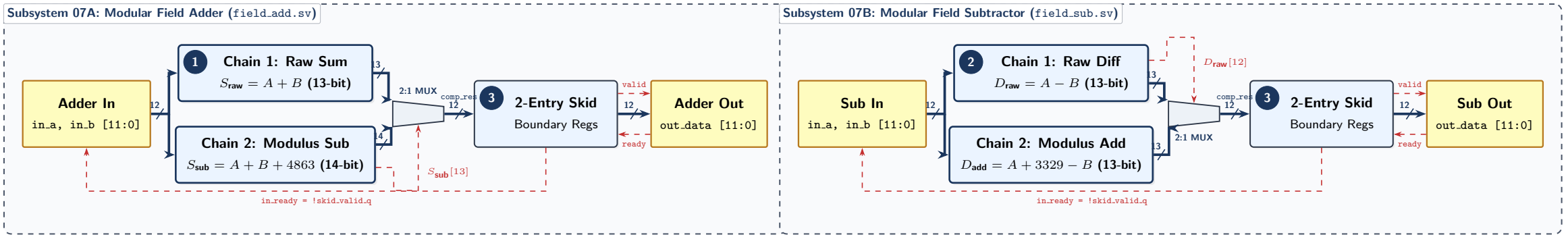
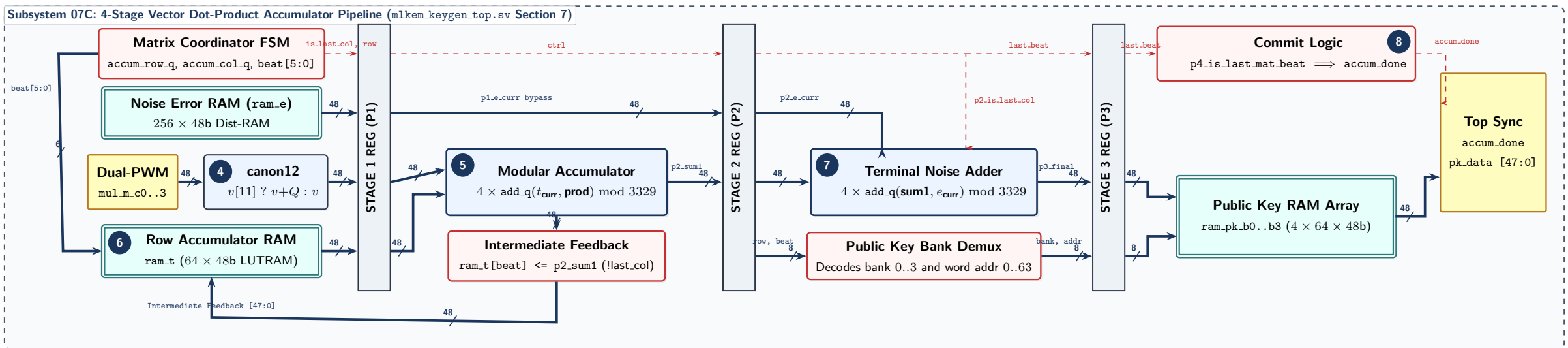
FIPS 203 ML-KEM Architecture: Modular Field ALU & Vector Accumulation Pipeline

Subsystem 07 Microarchitecture — Modular Arithmetic Units & Vector Accumulator Pipeline (`field_add.sv`, `field_sub.sv`, `mlkem_keygen_top.sv`) — Kintex UltraScale 450 MHz





ID	Component / Signal	Width / Format	Description
1	Fast Dual Carry Adder	12-bit / CARRY8	Evaluates $S_{\text{raw}} = A + B$ and $S_{\text{sub}} = A + B + 4863$ in parallel; bit 13 selects final sum via 2:1 MUX.
2	Fast Dual Borrow Subtractor	12-bit / CARRY8	Evaluates $D_{\text{raw}} = A - B$ and $D_{\text{add}} = A + 3329 - B$ in parallel; borrow bit 12 selects residue via 2:1 MUX.
3	2-Entry Elastic Skid Buffers	12-bit / Boundary Regs	Decouples handshakes (`in_ready = !skid_valid_q`); absorbs in-flight beats during stalls to maintain $II = 1$ at 450 MHz.
4	Quad Canonicalizer (`canon12`)	4×12b Bus	Normalizes multiplier quads from $[-1664, 1664]$ to $[0, 3328]$ via $v[11]\ ?\ v + 3329 : v$.
5	Stage 2 Quad Accumulator	4×12b Parallel ALUs	Computes running dot-product sums ($t_{\text{curr}} + \text{prod}$) mod 3329 across k matrix columns using parallel modular adders.
6	Row Accumulator RAM (`ram_t`)	64×48b LUTRAM	Distributed RAM holding intermediate polynomial vector sums; bypassed on col 0 and updated on non-final columns.
7	Stage 3 Terminal Noise Adder	4×12b Parallel ALUs	Adds sampled CBD noise vector $\mathbf{e}$ on terminal column ($j = k - 1$) and demultiplexes public key quads into 4 RAM banks.
8	Stage 4 Commit & Sync	Control Strobe	Registers final matrix beat (`p4_is_last_mat_beat`) and issues single-cycle `accum_done` strobe to Top Master FSM.